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Rietveld Refinement of Debye–Scherrer Synchrotron X-ray Data from Al_2O_3

BY P. THOMPSON, D. E. COX AND J. B. HASTINGS

Brookhaven National Laboratory, Upton, NY 11973, *USA*

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Abstract

The application of the Rietveld refinement technique to synchrotron X-ray data collected from a capillary sample of Al_2O_3 in Debye–Scherrer geometry is described. The data were obtained at the Cornell High Energy Synchrotron Source (CHESS) with an Si(111) double-crystal monochromator and a Ge(111) crystal analyzer. Fits to a number of well resolved individual peaks demonstrate that the peak shapes are very well described by the pseudo-Voigt function, which is a simple approximation to the convolution of Gaussian and Lorentzian functions. The variation of the Gaussian and Lorentzian half widths, Γ_G and Γ_L, with Bragg angle can be approximated quite closely by the functions $V \tan\theta$ and $X/\cos\theta$ which represent the contributions from instrumental resolution and particle-size broadening respectively. Rietveld refinement based on this model yields generally satisfactory results. The refined values of V and X are consistent with the expected vertical divergence ($\simeq 0{\cdot}1$ mrad) and the nominal particle size ($\simeq 0{\cdot}3$ μm). In particular, the use of a capillary specimen virtually eliminates preferred orientation effects, which are highly significant in flat-plate samples of this material.

Introduction

Since the introduction of the Rietveld profile-fitting method (Rietveld, 1969) for structural analysis of powder diffraction data collected using neutrons, there has been a growing interest in its application to X-ray diffraction (Malmros & Thomas, 1977; Young, Mackie & Von Dreele, 1977; Khattak & Cox, 1977; Wiles & Young, 1981; Young & Wiles, 1982; Will, Parrish & Huang, 1983). A number of methods have been tried and various diffraction geometries have been used, Guinier by Malmros & Thomas (1977), Bragg–Brentano by the other authors mentioned and, more recently, Debye–Scherrer (Thompson & Wood, 1983). Each has advantages and disadvantages regarding the application of the Rietveld method. For example, Bragg–Brentano and Guinier geometries give higher resolution and shorter data collection times than the simpler Debye–Scherrer method, but, apart from absorption, the geometrical and physical aberrations tend to be more complex and may be difficult to quantify over the full range of Bragg angles. Furthermore, particularly in Bragg–Brentano geometry, preferred orientation effects may be very significant. These are difficult to correct for and can systematically affect the deduced structure since they tend to be strongly correlated with the structure parameters, particularly the anisotropic thermal factors.

With the availability of dedicated synchrotron radiation sources, improvements in powder diffraction methods can be expected to lead to renewed interest in the Rietveld technique. It has already been demonstrated, for instance, that significant gains in resolution can be realized compared with that achievable on a conventional X-ray source [Thompson, Glazer, Albinati & Worgan, 1981; Buras & Christensen, 1981; Cox, Hastings, Thomlinson & Prewitt, 1983 (hereafter referred to as I); Hastings, Thomlinson & Cox, 1984 (hereafter referred to as II); Parrish, Hart & Huang, 1986; Christensen, Lehmann & Nielsen, 1985]. This alone would prove extremely useful for structure analysis of complex materials by the profile method. It is therefore of interest to apply the high resolution available with monochromatized synchrotron radiation to the simple diffraction geometry of the Debye–Scherrer method. The use of small rapidly rotating capillary specimens has many advantages over other methods; preferred orientation is greatly reduced, the diffraction geometry is simple, it is easier to construct environmental chambers, and only a small amount of material is needed, which is particularly useful if the specimen is reactive or moisture sensitive and must be encapsulated. The purpose of this paper is to demonstrate the feasibility of Rietveld refinement with data collected from a storage ring using a capillary specimen.

Experimental details

The experiments were performed on the high-energy synchrotron-radiation source at Cornell using the A3 station. The beam used was a focused beam from a two-crystal silicon monochromator diffracting in the vertical plane. The first was a flat rectangular Si(111) crystal 50 mm wide, 11 m from the source, which intercepted 0·14 mrad vertical by 4·4 mrad horizontal of the synchrotron radiation fan. Sagittal focusing over the 4·4 mrad of beam was achieved with a

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cylindrically bent triangular second Si(111) crystal (Batterman & Berman, 1983), giving a photon flux of approximately 10^{11} photons s^{-1} in a 4×2 mm beam with a spectral bandwidth of $\Delta\lambda/\lambda \simeq 10^{-3}$. A Philips diffractometer with the specimen position modified to hold a cylindrical Debye–Scherrer capillary was aligned in the synchrotron beam at the focal spot 3·9 m from the monochromator. The diffractometer arm was modified to hold a Ge(111) analyzer crystal in place of the standard pyrolytic graphite diffracted-beam monochromator used with a conventional source. Vertical Soller slits with a horizontal divergence of 1·6° [defined as in Klug & Alexander (1974)] were placed just before the analyzer crystal to limit the axial divergence of the scattered X-rays. The purpose of the analyzer crystal in this case is not just to reduce the background and eliminate fluorescence but also to replace the conventional receiving slit of the spectrometer by acting as a very narrow 'angular slit'. This allows much higher angular resolution and also virtually eliminates specimen displacement and eccentricity errors, since the X-rays entering the detector are independent of the point of origin (see II). As a result, the resolution is essentially independent of the diameter of the sample capillary and furthermore the profile shape will not be affected by absorption as in conventional Debye–Scherrer geometry.

The specimen used was Al_2O_3 crushed from a sintered pellet of a National Bureau of Standards reference material with a nominal particle size of 0·3 μm, and loosely packed in a 0·5 mm diameter capillary which was rotated at about 10 r min^{-1}. The data collection time extended from the middle of one fill of the storage ring through two subsequent fills, about 8 h in all. The wavelength was determined to be 1·4885 Å by calibration with a standard Ni sample just before the start of the experiment. From measurements of the weight and dimensions of the sample, the effective value of μR at this wavelength was found to be 0·9.

Data were collected by step-scanning at 0·01° intervals over a number of regions between 23 and 98° for a fixed monitor count provided by an ion chamber placed in the incident beam. The angular ranges covered were sufficiently wide to enable a reliable estimate of background to be made well away from the tails of the peaks. A typical collection time was around 5 s per point.

Results

Peak shape analysis

Several functions have been used to model X-ray peak shapes, and a recent review of these is given by Young & Wiles (1982). The Voigt function (convolution of Gaussian and Lorentzian functions) has been shown to give good results, particularly for the analysis of particle-size and strain broadening effects (Langford, 1978; Suortti, Ahtee & Unionius, 1979; de Keijser, Langford, Mittemeijer & Vogels, 1982). A simple approximation to the Voigt function which is much more convenient from the programming point of view is provided by the 'pseudo-Voigt' function (Wertheim, Butler, West & Buchanan, 1974; Young & Wiles, 1982), which has previously been used successfully in I and II for the analysis of synchrotron X-ray data from flat-plate samples. This is given by the expression

$$I(\Delta 2\theta) = I_0\{(2\eta/\pi\Gamma)[1 + 4(\Delta 2\theta/\Gamma)^2]^{-1} + (1-\eta)(2/\Gamma)(\ln 2/\pi)^{1/2} \times \exp[-4\ln 2(\Delta 2\theta/\Gamma)^2]\}, \quad (1)$$

where I_0 is the integrated intensity, $I(\Delta 2\theta)$ is the intensity of a point displaced by $\Delta 2\theta$ from the Bragg angle $2\theta_k$, Γ is the full width at half-maximum (FWHM) and η is a parameter which mixes the two functions. It relates the FWHM's of the individual Lorentzian (Γ_L) and Gaussian (Γ_G) components, and can be represented by a simple series expansion,

$$\eta = 1{\cdot}36603(\Gamma_L/\Gamma) - 0{\cdot}47719(\Gamma_L/\Gamma)^2 + 0{\cdot}11116(\Gamma_L/\Gamma)^3. \quad (2)$$

The coefficients used in this expression differ from those used earlier in II because of the normalization factors in (1).

Instead of using η and Γ as variables in the fitting procedure, it is better to use Γ_L and Γ_G directly, since η is difficult to relate to any physical parameters, while Γ_L and Γ_G can be readily identified with particle-size effects, instrumental resolution and possibly strain broadening. The approximation used for Γ is another simple series expansion derived from a set of computer-generated convolutions,

$$\Gamma = (\Gamma_G^5 + 2{\cdot}69269\Gamma_G^4\Gamma_L + 2{\cdot}42843\Gamma_G^3\Gamma_L^2 + 4{\cdot}47163\Gamma_G^2\Gamma_L^3 + 0{\cdot}07842\Gamma_G\Gamma_L^4 + \Gamma_L^5)^{1/5}. \quad (3)$$

In Fig. 1 is shown a plot of the 116 reflection, which is the strongest peak in the pattern. This could not be fitted adequately by either a Gaussian or Lorentzian peak shape, but the pseudo-Voigt function gave an excellent fit, as shown by the solid line and the difference plot below. All the stronger well resolved single peaks were fitted individually in this way with six variable parameters: I_0, $2\theta_k$, Γ_G, Γ_L and a linear sloping background defined by two parameters, over a range of 5Γ on either side of the peak. The values of the goodness-of-fit index χ^2 ranged between 1·0 and 1·5 with an average value of about 1·2. The background parameters were assigned initial values obtained by averaging well away from the peaks, and were not refined at first; in fact it made very little difference to the quality of the fit whether these parameters were varied or not.

The overall variation of Γ_G, Γ_L and Γ with 2θ is summarized in Fig. 2. The solid lines represent a least-squares fit of Γ_G and Γ_L to the one-parameter functions $V \tan\theta$ and $X/\cos\theta$ respectively. The latter term can be related to the Scherrer broadening for small particles, and the refined value of 0·035(1)° corresponds to a mean apparent particle size of about 0·25 μm, in reasonable agreement with the nominal value and previous results for a flat-plate sample of this material obtained in II with unfocused radiation.

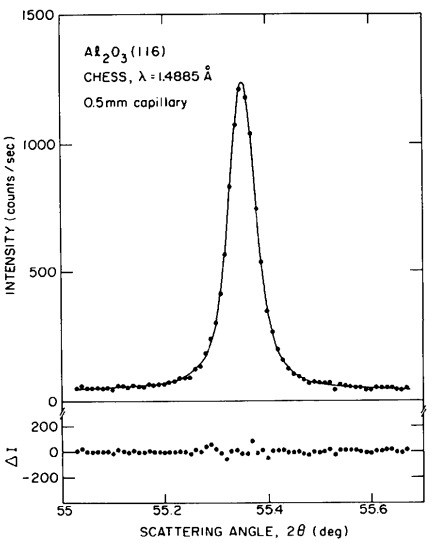


Fig. 1. Least-squares fit to experimental points for the 116 peak from Al_2O_3 based upon the pseudo-Voigt function described in the text with refined values $I_0 = 20864(241)$, $2\theta_k = 55{\cdot}3534(3)°$, $\Gamma_G = 0{\cdot}039(2)$, $\Gamma_L = 0{\cdot}040(2)°$. Goodness-of-fit $\chi^2 = 1{\cdot}24$. Difference plot is shown at bottom.

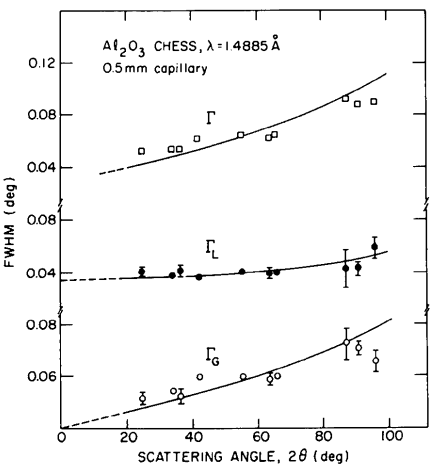


Fig. 2. Variation of Γ_G, Γ_L and Γ with 2θ for Al_2O_3. The solid lines for Γ_G and Γ_L represent least-squares fits to the functions $V \tan\theta$ and $X/\cos\theta$, with $V = 0{\cdot}069(5)$ and $X = 0.035(1)°$.

The $V \tan\theta$ term is analogous to the more general expression

$$\Gamma_G = (U \tan^2\theta + V \tan\theta + W)^{1/2}, \qquad (4)$$

familiar to neutron diffractionists (Rietveld, 1969). It has previously been shown in I and II that for synchrotron radiation the same resolution function can be applied, with the simplification that because of the high natural collimation of the synchrotron X-ray beam, the resolution is governed mainly by the vertical divergence of the latter and the monochromator and analyzer configuration. Neglecting the contribution from the Darwin widths of the crystals and the horizontal divergence, Γ_G is given by the expression

$$\Gamma_G = |\varphi(1 - \tan\theta_A/\tan\theta_M + 2\tan\theta/\tan\theta_M)|, \qquad (5)$$

where θ_M and θ_A are the Bragg angles of the monochromator and analyzer respectively. For the Si(111)–Ge(111) configuration used in this experiment, the constant term in (5) is negligibly small, and hence $V \simeq 2\varphi/\tan\theta_M$. The refined value of 0·069(5)° (Fig. 2) corresponds to $\varphi \simeq 0{\cdot}15$ mrad. This is somewhat larger than the estimated value for this sample geometry ($\simeq 0{\cdot}1$ mrad) and indicates the presence of some strain broadening. This is normally supposed to be Gaussian in character (Langford, 1978) and gives an additional contribution proportional to $2(\Delta d/d)\tan\theta$, where Δd is the spread in the d spacing due to small microstrains. This contribution is absorbed into the refined value of V.

Although the quality of the fit to the individual peaks described above is very good, some unexpected discrepancies $\Delta 2\theta_k$ between observed and calculated peak positions were noted. This is quite surprising because in previous experiments (II) very small discrepancies were observed, as indicated by a 'figure of

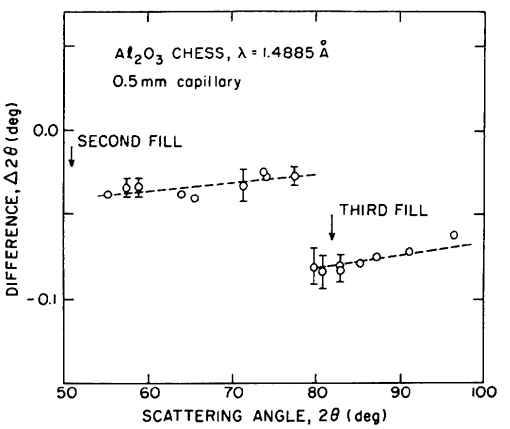


Fig. 3. Drift in $\Delta 2\theta_k (= 2\theta_{obs} - 2\theta_{calc})$ for Al_2O_3 data taken in two successive fills. $2\theta_{calc}$ values based on lattice parameters determined in II ($a = 4{\cdot}7584$, $c = 12{\cdot}9903$ Å) with $\lambda = 1{\cdot}4885$ Å. The broken lines represent the effect of a change in λ of 0·0005 Å over the duration of the fill. A discontinuous change in $\Delta 2\theta_k$ of about 0·05° at $2\theta = 78°$ is evident.

merit' [F_N, defined as $|\Delta 2\theta_k|^{-1}\ (N/N_{poss})$ by Smith & Snyder (1979)] of over 300. One type of discrepancy is illustrated in Fig. 3, *i.e.* a systematic gradual change in $\Delta 2\theta_k$ over the period of the fill. A possible explanation for this is a slight change in wavelength with decay of the stored beam, as indicated by the broken lines in Fig. 3, which show the effect of a change in wavelength of 0·0005 Å over the duration of the fill. This kind of effect was not observed in previous experiments using an unfocused beam in which the incident flux and horizontal beam divergence were much smaller. It is therefore possible that crystal heating effects and the monochromator servomechanism feedback might have led to slight angle changes.

In addition to these gradual changes, two discontinuous changes in $\Delta 2\theta_k$ were observed at about 50 and 80° for which no explanation comes readily to mind. The latter of these is shown in Fig. 3, and is seen to occur towards the end of the second fill. For the purposes of Rietveld refinement, allowance was made for these discontinuities by small adjustments to the zero-point parameter, $\Delta 2\theta_0$, in the regions 50–78° and 78–98°.

Rietveld refinement

Rietveld refinement requires a description of the intensity of the profile from each reflection contributing to a point on the diffraction pattern. The intensity y (corrected for background) at point i in the pattern can be computed from

$$y_i \propto \sum_k w_{ik} S_k^2, \tag{6}$$

where S_k contains the structural information and w_{ik} describes the profile of the kth reflection contributing at the point i. The non-structural term

$$w_{ik} \propto t j_k L_k f_{ik}(\Delta 2\theta_{ik}) \tag{7}$$

requires a detailed description of the peak shape $f_{ik}(\Delta 2\theta_{ik})$. In this expression t is the step interval of the counter, L_k is the Lorentz–polarization factor and j_k the multiplicity. Various functions to describe $f_{ik}(\Delta 2\theta_{ik})$ have been reviewed by Young & Wiles (1982), including the pseudo-Voigt approximation. As discussed above, not only does this account very well for the peak shapes observed in the present experiment, it also allows the variation in half-widths with 2θ to be incorporated in a simple and physically intuitive way through the expressions

$$\Gamma_G = V \tan\theta, \tag{8a}$$

$$\Gamma_L = X/\cos\theta. \tag{8b}$$

With this model, Rietveld refinement of the data yielded a quite satisfactory fit, as indicated by the first column of Table 1. Ten parameters were refined, including Al(z) and O(x), isotropic temperature factors and the two half-width parameters V and X. Background was estimated by linear interpolation between average values observed in regions well away from the peaks. The total number of reflections included was 30. Table 1 also shows the results of a second refinement in which a preferred-orientation correction was applied, with a correction term

$$I_{corr} = I_{obs} \exp[-G(\pi/2 - \varphi)^2] \tag{9}$$

in which G is a refinable parameter and φ is the angle between the scattering vector and the [001] axis. As shown in column II of Table 1, there is only a slight improvement in the agreement factors, and the correction term can be regarded as barely significant. The value of $G = 0{\cdot}033(7)$ can be compared to a much larger value of 0·195(9) obtained in a previous refinement of data from a similar Al_2O_3 sample collected with Cu $K\beta$ radiation with Bragg–Brentano flat-plate geometry (column III, Table 1).

Table 1. *Rietveld refinement of* Al_2O_3 *data*

Columns I and II: synchrotron data, 0·5 mm capillary. III: Cu $K\beta$ data, flat-plate (Cox, Moodenbaugh, Sleight & Chen, 1980). IV: Mo $K\alpha$ data, single crystal (Calvert, Gabe & Le Page, 1981). Figures in parentheses are e.s.d.'s referred to the least significant digit. R factors are as defined by Rietveld (1969) and Young, Prince & Sparks (1982). χ^2 is the goodness-of-fit index [$= (R_{wp}/R_E)^2$].

	I	II	III	IV
Al:z	0·3518(1)	0·3518(1)	0·35200(7)	0·35219(1)
Al:B(Å^2)	0·68(5)	0·68(5)	0·40(1)	0·26(1)
O:x	0·6918(6)	0·6914(6)	0·6936(3)	0·69367(5)
O:B(Å^2)	0·71(7)	0·74(7)	0·51(2)	0·28(1)
V(°)	0·059(2)	0·059(2)	–	–
X(°)	0·037(1)	0·037(1)	–	–
G(rad^{-2})	–	0·033(7)	0·195(9)	–
a(Å)	4·7586(1)	4·7586(1)	4·75855(2)	4·75999(3)
c(Å)	12·9897(1)	12·9897(1)	12·9906(1)	12·99481(7)
R_I	0·066	0·065	0·038	–
R_{wp}	0·222	0·220	0·149	–
R_E	0·144	0·144	0·111	–
χ^2	2·38	2·33	1·80	–

The refined positional parameters listed in Table 1 are in reasonable agreement with the single-crystal values (column IV), but the temperature factors are about a factor of 2·5 larger. A similar, although less pronounced, trend is apparent with the Cu $K\beta$ data, and at present we do not understand the reason for this discrepancy. Adjustments to the background had little effect, and the inclusion of an absorption correction would lead to a slight increase of about 0·2 Å^2 in the refined values (Hewat, 1979).

The refined values of the half-width parameters V and X are in good agreement with those obtained from the individual peak fits in Fig. 2, and confirm that sample effects and instrumental resolution can be incorporated into Rietveld refinement in a simple and physically plausible way rather than by the introduction of quite arbitrary and empirical functions.

In conclusion, it appears that the Debye–Scherrer method coupled with a perfect-crystal analyzer should

be capable of providing Rietveld refinements of high accuracy. In comparison with flat-plate techniques, the simple geometry may offer some important advantages; only a small amount of material is required, it is well suited for reactive specimens which must be encapsulated, preferred-orientation effects are greatly reduced, and it is easier to construct environmental chambers. However, the present experiments indicate that temperature control and monochromator stability should be given careful consideration, particularly if focused optics are used.

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